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MILITARY SPECIFICATION

RINGER, TELEPHONE TA-48/FT

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers one type of telephone ringer designated as Ringer TA-48/FT, which converts a 60-cycle supply to a 20-cycle ringing current.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, and drawings of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

QQ-S-571 — Solder, Soft (Tin, Tin-Lead, and Lead-Silver).

PPP-B-621 — Boxes, Wood, Nailed and Lock-Corner.

PPP-B-636 — Boxes, Fiber.

PPP-T-60 — Tape; Pressure Sensitive Adhesive, Waterproof — for Packaging and Sealing.

MILITARY

JAN-I-10 — Insulating Materials, Ceramic Radio, Class L.

MIL-M-14 — Molding Plastics and Molded Plastic Parts, Thermosetting.

MIL-C-25 — Capacitor, Fixed, Paper-Dielectric, Direct-Current (Hermeticaly Sealed in Metallic Cases).

MIL-T-27 — Transformers and Inductors (Audio, Power, and Pulse).

MIL-W-76 — Wire and Cable, Hook-Up, Electrical, Insulated.

MIL-P-79 — Plastic Materials, Laminated, Thermosetting Rods and Tubes.

MIL-P-116 — Preservation, Methods of.

FSC 5805

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MIL-B-117 — Bags, Interior, Packaging.

MIL-V-173 — Varnish, Moisture- and Fungus-Resistant, for the Treatment of Communications, Electronic and Associated Electrical Equipment.

MIL-I-631 — Insulation, Electrical, Synthetic-Resin Composition, Nonrigid.

JAN-A-669 — Anti-seize Compound, White Lead Base, General Purpose (For Threaded Fittings).

MIL-T-713 — Twine and Tape, Lacing and Tying (For use in Electrical and Electronic Equipment).

MIL-P-997 — Plastic-Material, Laminated, Thermosetting, Electrical-Insulating: Sheets, Glass Cloth, Silicone Resin.

MIL-R-3065 — Rubber and Synthetic Rubber Compounds; General Purpose (Except Tires, Inner Tubes, Sponge Rubber and Hard Rubber).

MIL-P-3115 — Plastic-Material, Laminated, Thermosetting, Sheets, Paper-Base, Phenolic-Resin.

MIL-P-3408 — Plastic-Material, Molding; Rigid Thermoplastic, Aniline Formaldehyde; For use in Electronic, Communications, and Allied Electrical Equipment.

MIL-P-3409 — Plastic-Material, Molding; Rigid Thermoplastic Polydichlorostyrene; For use in Electronic, Communications, and Allied Electrical Equipment.

MIL-P-3410 — Plastic-Material, Molding; Rigid Thermoplastic, Polyvinyl Chloride and Copolymers Thereof; For use in Electronic, Communications, and Allied Electrical Equipment.

MIL-P-3411 — Plastic-Material, Molding; Rigid Thermoplastic, Vinylidene Chloride; For use in Electronic, Communications, and Allied Electrical Equipment.

MIL-P-3412 — Plastic-Material, Molding; Rigid Thermoplastic, Ethyl Cellulose; For use in Electronic Communications, and Allied Electrical Equipment.

MIL-P-3413 — Plastic-Material, Molding; Rigid Thermoplastic, Po-

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lystyrene; For use in Electrical Communications, and Allied Electrical Equipment.

MIL-P-3414 — Plastic - Material, Molding; Rigid Thermoplastic, Cellulose Acetate Butyrate; For use in Electronic, Communications, and Allied Electrical Equipment.

MIL-P-3415 — Plastic - Material, Molding; Rigid Thermoplastic, Cellulose Acetate; For use in Electronic Communications, and Allied Electrical Equipment.

MIL-P-3416 — Plastic - Material, Molding; Rigid Thermoplastic, Acrylic; For use in Electronic, Communications, and Allied Electrical Equipment.

MIL-C-3432 — Cable, Flexible and Extra Flexible, 300 and 600 Volts.

MIL-W-3861 — Wire, Electrical (Bare Copper).

MIL-I-3930 — Insulating and Jacketing Compounds, Electrical (For Cable, Cord, and Wire).

MIL-I-7798 — Insulating Tape, Electrical, Pressure Sensitive Adhesive, Plastic.

MIL-T-10186 — Tabular List of Parts for Signal Equipment.

MIL-R-10187 — Reference Symbols for Signal Corps Equipment.

MIL-F-14072 — Finishes for Ground Signal Equipment.

MIL-P-15035 — Plastic - Material, Laminated, Thermosetting; Sheets, Cotton-Fabric-Base, Phenolic-Resin.

MIL-P-15037 — Plastic - Material, Laminated, Thermosetting, Sheets, Glass-Cloth Melamine-Resin.

MIL-P-15047 — Plastic - Material, Laminated, Thermosetting, Sheets, Nylon Fabric-Base, Phenolic-Resin.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-109 — Inspection Terms and Definitions.

MIL-STD-129 — Marking for Shipment and Storage.

MIL-STD-130 — Identification Marking of U. S. Military Property.

MIL-STD-169 — Extreme — Temperature Cycle.

MIL-STD-170 — Moisture Resistance Test Cycle, For Ground Signal Equipment.

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MIL–STD–252 — Wired Equipment, Classification of Visual and Mechanical Defects.

DRAWINGS

SIGNAL CORPS

SC–D–17269 — Caution Plates.

SC–DL–135529 — Ringer TA–48/FT.

(Copies of specifications, drawings, and standards required by contractors in connection with specified procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and identifying number of symbol should be stipulated when requesting copies.)

2.2 Other publications. The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 — Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Description. Ringer TA–48/FT is a static type ringer, housed in a metal cabinet, designed to supply ringing current for common battery manual telephone switchboards, converting 60-cycle 115-volt supply to a 20-cycle ringing current. Ungrounded ringing current or divided ringing to ground for party ringing, can also be delivered. The ringer is designed for mounting on central office switchboard frame or on a wall.

3.2 Preproduction samples. The contractor shall furnish preproduction samples for approval as required by the invitation for bids and contract (see 6.2 (b) and (c)).

3.3 Construction. The equipment shall be constructed in accordance with drawings SC–DL–135529—Ringer TA–48/FT.

3.4 Parts, materials and processes. All parts, materials and processes not specifically mentioned herein shall be in accordance with the applicable specifications and drawings. The selection of class, grade or type of material shall be such that, when mounted in the equipment, the part or material will perform its intended function under the service conditions specified in 3.11.

3.4.1 *Insulation material.* Insulation material shall not support rapid combustion. The material used shall be sufficiently rigid and free from cold flow to maintain its function during exposure of the equipment to the service conditions specified in 3.11.

3.4.2 *Plastic material.* Where not machined, plastic material shall have the original smooth or polished surfaces. Surfaces that have been machined shall be as smooth as practicable in accordance with good manufacturing practice for the kind of material used and the intended application.

3.4.2.1 *Thermosetting.* Thermosetting plastic material shall conform to table I.

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TABLE I. *Thermosetting plastic materials*

Specification		
Number	Title	Types
MIL–M–14	Molding, Plastics and Molded Plastic Parts, Thermosetting.	CFG, CFI–5, CFI–20, CMG, CMI–5, MFE, MME.
MIL–P–79 [1]	Plastic-Materials, Laminated, Thermosetting; Rods and Tubes.	*All types.*
MIL–P–997	Plastic-Material, Laminated, Thermosetting; Electrical-Insulating; Sheet, Glass Cloth. Silicone Resin.	GSC
MIL–P–3115	Plastic-Material, Laminated, Thermosetting; Sheets, Paper Base, Phenolic-Resin.	PBE, PBG, PBE–P.
MIL–P–15035 [1]	Plastic-Material, Laminated, Thermosetting; Sheets, Cotton-Fabric-Base, Phenolic-Resin.	FBI, FSE, FBM, FBG
MIL–P–15037	Plastic-Material, Laminated, Thermosetting; Sheets, Glass-Cloth, Melamine-Resin.	GMG.
MIL–P–15047	Plastic-Material, Laminated, Thermosetting; Sheets, Nylon-Fabric-Base, Phenolic-Resin.	NPG.

[1] Cotton fabric base laminates shall not be used unless otherwise specified or unless high impact resistance is required for gears, cams, or similar parts.

3.4.2.2 *Thermoplastic insulation, nonrigid.* Nonrigid thermoplastic electrical insulation shall conform to Specification MIL–I–631.

3.4.2.3 *Thermoplastic, rigid.* Thermoplastic rigid molding material shall conform to the following specifications:

MIL–P–3408— Aniline Formaldehyde.

MIL–P–3409— Polydichlorostyrene.

MIL–P–3410— Polyvinyl Chloride and Copolymers Thereof.

MIL–P–3411— Vinylidene Chloride.

MIL–P–3412— Ethyl Cellulose.

MIL–P–3413— Polystyrene.

MIL–P–3414— Cellulose Acetate Butyrate.

MIL–P–3415— Cellulose Acetate.

MIL–P–3416— Acrylic.

3.4.3 *Ceramic material.* Ceramic material, except when used as the dielectric in capacitors, shall conform to Specification JAN–I–10.

3.4.4 *Rubber and synthetic rubber.* Rubber and synthetic rubber used for general mechanical purposes shall be in accordance with Specification MIL–R–3065.

3.4.5 *Solder, soft.* Soft solder shall be in accordance with Specification QQ–S–571, and where used for electrical connections shall have a minimum tin content of 39.0 percent by weight.

3.4.5.1 *Soldering flux and cleaning agents.* Flux for soldering shall be rosin or rosin and alcohol. The rosin shall be grade "WW" (water white) or better. No acid or acid salts shall be used in preparation for or during soldering. Where soldered materials are such that the use of alcohol and rosin is impracticable, the contractor shall secure permission from the bureau or agency concerned before using alternate fluxes or cleaning agents. When use of alternate materials is allowed, such materials shall be chemically neutralized and remove immediately afterward.

3.4.5.2 *Soldering.* Soldered connections

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shall be neat. There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting action, and shall not be crystallized, overheated, or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Wherever practicable, excess rosin shall be removed by the use of a wire brush and then a dry cloth; any resulting loose flakes of rosin shall be carefully removed from the inside of the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.4.6 *Welding.* Surfaces to be welded shall be cleaned in accordance with good commercial practice to be free from rust, scale, paint, grease, and other foreign material. Where embrittlement of the parent metal is likely to result, preheating or annealing shall be employed. Welds shall be of ample size with thorough penetration and good fusion and shall be free from scabs, blisters, abnormal pockmarks, bruises, or other harmful defects. Welds shall be secure and free from brittle metal and the welded material shall retain sufficient residual strength for the intended purpose. Where necessary, the full strength of the joined parts shall be developed. (Some weakening of welded material may be expected 2 to 3 inches from the weld because of annealing of the material as a result of the welding operation.) Where undesirable internal stresses may result from welding, the welded assembly shall be annealed to relieve the stresses. Welds shall be cleaned of scale, oxidation products, and excess of flux by sandblasting, wire brushing, or other suitable means. Acid used for cleaning shall be completely neutralized and removed. Electrodes used for arc welding shall be of a type that will produce a weld having chemical and physical properties similar to those of the parent metal. Where tack or spot welds are employed to secure parts permanently, the number of welds shall be sufficient to provide adequate strength and at least two welds shall be used to hold each part.

3.4.6.1 *Cleaning.* Metal parts, after fabrication, shall be cleaned in accordance with good commercial practice. Cleaning processes shall have no visible or latent, deleterious effect on the equipment. Corrosive materials shall be removed completely before parts are assembled into the equipment. After assembly, components shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. If necessary, such cleaning shall also be performed before final assembly of components.

3.5 **Wiring and cabling.** Wiring and cabling shall be neat and sturdy.

3.5.1 *Cabling.* Insulated wires shall be formed into cables, except where operation of the equipment would be adversely affected thereby or where it is mechanically impracticable as in the case where the resulting cables would be excessively large and would interfere with operation or maintenance of the equipment. The cabling of wires shall be effected by twine conforming to Specification MIL–T–713; or by vinylchloride-acetate backed, pressure-sensitive tape, type VF, conforming to Specification MIL–T–7798, except that the tape shall be natural color.

3.5.1.1 *Slack.* Wires and cables shall be as short as practicable except that sufficient slack shall be provided to (a) prevent undue stress on cable forms, wires, and connections, (b) enable parts to be removed and replaced during servicing without disconnecting other parts, and (c) facilitate field repair of broken or cut wires.

3.5.1.2 *Protection.* Wires and cables shall be so placed and protected as to avoid contact with rough or irregular surfaces or sharp edges, under specified service conditions. Wires shall not be bent sharply

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where they enter insulation material. Where wires run through holes in metal partitions, shields, and the like, they shall be protected by suitable grommets or bushings.

3.5.1.3 *Connections.* Soldering leads, lugs, and terminals shall be tinned, silver-plated, or lead-alloy coated. Wires subject to breakage at the connections shall be provided with terminals that grip the wire insulation. Solderless terminals may be used on tinned, stranded wire in power or audio circuits, but other wire connections shall be soldered. Where practicable, wires soldered to terminals shall be looped at least once and not more than twice around the terminals before soldering, or equivalent means shall be employed. Textile insulation ends of wires shall be secured against fraying by mechanical means or by application of varnish or lacquer conforming to Specification MIL–V–173.

3.5.2 *Wire, hookup.* Wherever practicable, hookup wire shall be stranded. The cross-sectional area of the conductor shall be sufficient to carry the necessary current with a minimum of overheating, such that the service life of the equipment will not be shortened by electrical or mechanical deterioration of the wire from that cause. Hookup wire shall be in accordance with Specification MIL–W–76.

3.5.3 *Cable and cable assemblies.*

3.5.3.1 *Power and audio frequency.* The conductor material of power and audio frequency cable shall be tinned, stranded, copper wire conforming to Specification MIL–W–3861.

3.5.3.2 Power or audio frequency cable or cable assemblies shall be insulated and jacketed with Buna S in accordance with Specification MIL–I–3930. Where practicable, power cable or cable assemblies shall also conform to Specification MIL–C–3432.

3.6 Dielectric strength. Dielectric strength shall meet the tests of 4.5.

3.7 Screws, nuts, and other threaded parts. Screw threads shall be in accordance with Handbook H28, and shall be class 2 fit unless otherwise specified. Wherever practicable, screw threads shall be commercial standard sizes. Threads in soft material shall be in the American National coarse thread or fine thread series, as suitable for the particular application. (Small screws, except where used in soft material, usually require fine threads in order to withstand vibration.)

3.7.1 *Securing of threaded parts.* Threaded parts shall be properly secured to withstand specified vibration, bounce, and shock tests without loosening. If retaining compound is used in addition to other means of securing a threaded part, the part shall be capable of withstanding the tests before the compound is applied. Except where disassembly will not be necessary during field use of the equipment or its maintenance, threaded parts shall be capable or repeated disassembly and reassembly. Wherever practicable, each threaded part shall be secured by one or more of the following means: lockwashers, locknuts, stopnuts, cotter pins, safety wires, retaining compound, or staking. Retaining compound shall not be solely depended upon where parts will be disassembled and reassembled. Where nonmetallic surfaces are involved, apply the compound on all bearing surfaces (under the screw head, on the washers, and on the nut). Staking shall not be used where the parts will be disassembled and reassembled.

3.7.2 *Antiseize compound.* Antiseize compound complying with Specification JAN–A–669 shall be used to prevent binding of threads where one or both of the threaded surfaces are of aluminum or magnesium.

3.8 Ease of adjustment. All controls shall operate easily and smoothly without excessive motion, binding, scraping or cutting. A

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small amount of lubricant (graphite or equivalent) shall be applied if it does not interfere with operation.

3.9 Safety. With the equipment assembled and set up for operation, personnel shall be protected from contact with potentials in excess of 30 volts to ground chassis, or frame, including potentials on charged capacitors.

3.10 Interchangeability. Corresponding components, replaceable subassemblies, and replaceable parts on contract shall be physically and functionally interchanegable as units without modification thereof or of other items with which the units are used. When dimensions, ratings, characteristics, etc., are not specified, the manufacturer's design limits shall be used to determine compliance with the foregoing. If the contractor is in doubt as to whether a particular subassembly or part is to be considered replaceable, the contracting officer shall be consulted.

3.11 Temperature and humidity. The equipment shall meet the performance requirements of this specification while subjected to any of the following conditions:

3.11.1 *Temperature.* The equipment shall be capable of withstanding the test specified in 4.8.1 without any deleterious effect on the material or reduction in performance of the equipment.

3.11.2 *Moisture resistance test.* The equipment shall be subjected to five continuous cycles of the test as shown on MIL–STD–170.

3.12 Lacquer. All lacquer and varnish used to protect labels, reference symbols and other markings shall be in accordance with Specification MIL–V–173 and shall be such that the markings remain legible after application of varnish or lacquer.

3.13 Tools and/or running spare parts. Tools and/or running spare parts shall be furnished as specified (see 6.2(d)). Running spare parts shall be identical to corresponding integral parts of the equipment.

3.14 Marking.

3.14.1 *General.* Components and parts to which Army-Navy or Signal Corps nomenclature has been assigned shall bear identification marking in accordance with Standard MIL–STD–130.

3.14.2 *Marking of standard parts.* Parts covered by standard specifications shall be marked as required by those specifications.

3.14.3 Other parts not covered by standard specifications, including mechanical parts, shall be marked wherever space permits with the manufacturer's name or trademark, catalog or other identification number, value and rating (when practicable) and other pertinent information. When space is limited, as much as practicable of the required information shall be included, giving preference to the specified order. Subassemblies that are replaceable as a unit shall be marked in the same manner, but the marking shall be such as not to be confused with marking of any replaceable part.

3.14.4 *Reference symbols.* Reference symbols, assigned in accordance with Specification MIL–R–10187 shall be marked on the equipment to identify the various parts. The same reference symbol shall be used on the equipment, diagrams, labels, and in tabular list of parts (Specification MIL–T–10186). Appropriate reference symbol by which a part is designated shall be marked on a surface adjacent to the part, but when space is not available, the reference symbol shall be marked on the part itself. Spare parts shall not be marked with reference symbols.

3.14.5 *Visibility.* Reference symbols shall be so located that they are readily visible with the minimum removal of cases, covers, shields

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or adjacent parts. Wherever practicable, parts shall be so mounted that their marking will be readily visible with the minimum disassembly of the equipment.

3.14.6 *Fuses.* Reference symbols shall not be marked on fuses. The reference symbol of the fuse shall be marked adjacent to the fuse holder. When space is limited, the reference symbol of the fuse holder may be omitted to provide space for the reference symbol of the fuse.

3.14.7 *Method of application.* Nomenclature markings shall be applied in accordance with Standard MIL–STD–130. Markings on standard parts shall be aplied in accordance with the standard specifications covering the parts. Other markings specified in 3.14.3 and 3.14.4 shall be applied in a legible and permanent manner—

Molding.

Engraving.

Etching.

Steel stamping.

Die stamping.

Rubber stamping[1].

Silk screening[1].

Decalcomania[1].

Glyceryl-phthalate paint[1].

Permanent ink[1].

Stencilling[1].

[1] These markings shall be protected by a coat of clear lacquer or varnish complying with Specification MIL–V–173, except that markings applied with lacquer base material need not be so protected when the material meets the applicable requirements of that specification.

3.15 Technical manual caution notice. A technical manual caution notice in accordance with Drawing SC–D–17269, shall be mounted on the front panel.

3.16 Facsimiles. Before painting the labels, the contractor shall receive approval of facsimiles thereof, submitted by him to the bureau or agency concerned.

3.17 Capacitors. Fixed paper dielectric capacitors in metallic cases shall conform to Specification MIL–C–25.

3.18 Power transformers and inductors. Power transformers and inductors shall be grade 1 in accordance with Specification MIL–T–27. The class of each transformer or inductor shall be such that its maximum safe operating temperature will not be exceeded when the equipment is subjected to specified service conditions (see 3.11).

3.19 Mechanical requirements.

3.19.1 *Weight.* The weight of the ringer shal not exceed 35 pounds.

3.19.2 *Size.* The dimensions of the ringer shall not exceed 9⅝ inches wide by 5¾ inches high by 14⅛ inches long.

3.19.3 *Finish.* The equipment shall be given a gray enamel finish in accordance with Specification MIL–F–14072.

3.19.4 *Bounce.* The equipment shall meet full specification performance. There shall be no physical damage except surface abrasion (see 4.4).

3.20 Electrical requirements. The unit shall be capable of a sustained power output within the voltage range and at the frequency specified for any condition from no load to full load when connected to the power source specified.

3.20.1 *Power source.* The ringer shall operate from single phase a.c. power sources of 105 to 125 volts, 60 c.p.s.

3.20.2 *Power input.* The no-load input power and the full load input power shall not exceed 35 and 60 watts, respectively.

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3.20.3 *Power output.* The ringer shall deliver not less than 15 volts at 75 volts + 0 —10 percent and 20 c.p.s.

3.20.4 *Output frequency.* The output frequency shall be 20 c.p.s. for an input frequency of 60 c.p.s.

3.20.5 *Output voltage.* Ringer TA–48/FT shall have an output potential of 90 ± 5 volts at no-load. Voltage regulation from full load to no-load shall not exceed 10 percent.

3.20.6 *Moving parts.* The ringer shall have no vibrating parts. The only moving part shall be a suitable relay to start operation.

3.20.7 *Fuse.* A readily accessible 3-ampere plug fuse shall be provided in the line side of the ringer.

3.20.8 *Tone coil.* A tone coil shall be incorporated in the ringer to provide a reverting tone.

3.20.9 *RF interference.* The maximum allowable radio interference expressed in microvolts per kilocycle of band width (UV/KC) or mocrovolts per megacycle of band width (UV/MC) shall not exceed the following levels:

Radiated interference

Frequency (MCS)	*Permissible level*
0.35 to 40.0	0 – 25 UV/KC.
40.0 to 90.0	100 UV/MC.
90.0 to 1000.0	200 UV/MC.

Conducted interference to power l'nes

Frequency (MCS)	*Permissible level*
0.35 to 3.0	10 UV/KC.
4.0 to 40.0	15 UV/KC.

3.20.10 *Dielectric strength.* There shall be no evidence of breakdown or degradation in performance as a result of the test specified in 4.5.

3.21 **Workmanship.** The equipment shall be manufactured and assembled in a thoroughly workmanlike manner, in accordance with the applicable portions of the following paragraphs herein:

3.4.2 — Plastic material.

3.4.5.1 — Soldering flux and cleaning agents.

3.4.5.2 — Soldering.

3.4.6 — Welding.

3.5.1 — Cabling.

3.7.1 — Securing of threaded parts.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection, general.

4.1.1 *Definitions.* Standard MIL–STD–109 applies for definitions of inspection terms used herein.

4.1.2 *Classification of inspection.* Examination and testing of the ringers shall be classified as follows:

(a) Preproduction inspection.

(b) Acceptance inspection.

4.2 **Preproduction inspection.** Preproduction inspection shall consist of the group A inspection specified in table III and the preproduction inspection specified in table II. Other nondestructive inspection on preproduction samples may be performed to determine compliance with specified requirements.

TABLE II. *Preproduction inspection*

Inspection	Requirement paragraph	Inspection paragraph
Low temperature	3.11	4.8.1
High temperature	3.11	4.8.1
Humidity	3.11	4.8.2

4.3 **Acceptance inspection.** Acceptance inspection shall consist of a group A inspection.

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4.3.1 *Group A inspection.* Group A inspection shall consist of the examination and tests specified in table III. Statistical sampling and inspection shall be in accordance with Standard MIL-STD-105.

4.2.1 *Preproduction inspection of preparation for delivery.* This inspection shall consist of the group B inspection specified in table IV and the rough handling test specified in 5.4.1.

TABLE III. *Group A inspection*

Inspection	Requirement paragraph	Inspection paragraph	AQL	
			Major	Minor
Visual and mechanical				
Workmanship	3.21	4.6.5	1.5 dphu for group.	6.5 dphu for group.
Weight	3.19.1	4.6.2		
Size	3.19.2	4.6.3		
Finish	3.19.3	4.6.4		
Electrical				
Dielectric strength	3.6	4.5	1.5 dphu for group.	
Power input	3.20.2	4.7.3		
Power output	3.20.3	4.7.3		
Output voltage	3.20.5	4.7.1		
Output frequency	3.20.4	4.7.2		

4.3.2 *Acceptance inspection of preparation for delivery.* This inspection shall include group B inspection as specified in 4.3.2.2 through 4.3.2.3; no group A or group C inspection is required.

4.3.2.1 *Group B inspection of preparation for delivery.* This inspection (including sampling) shall conform to table IV and to the appendix to Standard MIL-STD-105. Inspection Level L-8 shall be used for normal inspection and L-6 for reduced inspection. Normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R-1.

4.3.2.2 *Inspection lot.* A lot for inspection of packaging shall be all unit packages of a particular method or similar combination of methods, submitted at one time. A lot for visual inspection of the pack shall be all completed packs which are identical and are submitted for inspection at one time.

4.3.2.3 *Procedure in case of failure.* If an inspection lot is rejected the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken.

4.3.2.1 *Conforming group B sample units.* Sample units which have been inspected for preparation for delivery shall be repackaged and repacked to conform to the requirements of section 5.

4.3.3 *Disposition of nonconforming product.* Disposition of lots and sample units rejected during group A or B inspection shall be in accordance with Standard MIL-STD-105.

4.4 Bounce preconditioning. The equipment, with shock mounts (if any) removed or blocked, shall be placed in its normal operating position on the table of the package tester as made by the L.A.B. Corporation, Summit, New Jersey, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a $\frac{1}{32}$-inch thick strip of material under one corner or edge of the equipment to a distance of 3 inches as the equipment bounces. The equipment shall be subjected to this preconditioning for 1 minute. After bounce precondition-

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TABLE IV. *Group B inspection of preparation for delivery*

Inspection	Requirement paragraph	Inspection paragraph	AQL
Packaging			
Visual inspection, minor	5.4	4.10	10%
Visual inspection, major	5.4	4.10	4%
Packing			
Visual inspection, minor	5.4	4.10	10%
Visual inspection, major	5.4	4.10	4%

ing, the equipment shall not be repaired, alined, cleaned or otherwise changed prior to subjection to acceptance inspection.

4.5 Dielectric strength test. The insulation strength of all circuits shall be tested to withstand 400 volts, or two times the maximum peak operating voltage, whichever is greater. The test voltage shall be d.c. or peak a.c. of the same value, and shall be applied for 10 seconds between wiring and adjacent metal noncurrent carrying parts.

4.6 Mechanical tests.

4.6.1 *Visual and mechanical inspection.* Equipment shall be examined for the defects listed in Standard MIL–STD–252.

4.6.2 *Weight.* Weight shall conform with mechanical requirements paragraph 3.19.1 of this specification.

4.6.3 *Size.* Size shall conform with mechanical requirements paragraph 3.19.2 of this specification.

4.6.4 *Finish.* Finish shall conform with mechanical requirements paragraph 3.19.3 of this specification.

4.6.5 *Workmanship.* Workmanship shall conform with requirements paragraph 3.21 of this specification.

4.7 Operational tests.

4.7.1 *Measurement of output voltage.* The ringer shall be connected to an input source of 115 volts 60 c.p.s. and the voltages measured at the two outputs (terminals 1 and 2 and 1 and 3) shall conform to 3.20.5. A load of 20 watts shall be applied to the output terminals and the voltages again measured for compliance with the regulation requirements of 3.20.5.

4.7.2 *Output frequency.* The output frequency of the ringer shall be tested as follows: The output of the ringer shall be applied to an oscilloscope. The wave shape shall approximate a sine wave. A 60 cycle voltage shall be applied to the horizontal deflection plates, and the ringer output to the vertical deflection plates. The resulting Lissajous figures shall indicate a 20 cycle output of the ringer.

4.7.3 *Power input.* The no-load input power and the full load input power shall conform to paragraph 3.20.2 of this specification.

4.7.4 *Power output.* The ringer shall conform to the requirement as set forth in paragraph 3.20.3 of this specification.

4.8 Service conditions test. The following tests will be performed on the preproduction samples at a Government laboratory.

4.8.1 *Low temperature and high temperature tests.* These tests shall conform with Standard MIL–STD–169, extreme temperature cycle. The tests specified in 4.5 and 4.7

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shall be performed at steps 3 and 8 of the temperature cycle.

4.8.2 *Humidity test.* One preproduction sample shall be subjected to the following test. When the contract requires that production samples be forwarded to a Government laboratory for environmental and life test, such samples shall be selected from the stipulated portions of production and shall be marked and shipped by the contractor as required by the contract (see 6.2(c). Such production samples shall be subjected to the following test.

4.8.3 *Moisture-resistance test for equipment.* The equipment shall be tested as follows.

4.8.3.1 *Test conditions.*

(a) Do not move equipment from the humidity chamber for measurements.

(b) Start measurements not more than 5 minutes after power is applied to the equipment. Complete measurements as rapidly as possible. Do not leave power on after measurements have been completed.

(c) Test sealed equipment with the seal intact except during last two cycles. Bring out leads for electrical measurements through a hole in the equipment enclosure and seal against entrance of moisture. Perform last two cycles with the chassis removed from its enclosure.

4.8.3.2 *Test method.*

(a) Dry at 130° ± 5° F. for 24 hours.

(b) Condition at 77° ± 5° F. and 40 to 50 percent relative humidity for 24 hours.

(c) Take measurements as specified in 4.8.3.3 and readjust or realine as necessary to meet full specification requirements.

(d) Subject to continuous cycling for five 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to Standard MIL–STD–170. The measurements specified in 4.8.3.3 shall be made during the periods shown on the standard and shall comply with 4.8.3.3.

(e) After cycling has been completed, condition the equipment for 24 hours at 77° ± 5° F. and 40 to 60 percent relative humidity. Then adjust for optimum performance, using only those means provided by the equipment. No repair or replacement of parts shall be made. After adjustment, the equipment shall meet full specified requirements of paragraph 4.8.3.3.

4.8.3.3 *Performance.* The tests specified in 4.5 and 4.7 shall be performed at the designated phase in the test cycle. There shall be no mechanical deterioration, as evidenced by corrosion of metal parts, or binding of moving parts, and there shall be no degradation in performance from that obtained prior to the test cycles.

4.8.3.4 *Failure.* If any equipment fails to meet the performance specified in 4.8.3.3 during cycling, it shall be realined or readjusted once. If the equipment then fails to meet 4.8.3.3 or fails subsequently during cycling, it does not pass the test. In addition, if the equipment fails to meet full specification requirements after conditioning and adjustment, as specified in 4.8.3.2 (e), it does not pass the test.

4.9 Inspection for interchangeability. The dimensions listed below shall be gauged or

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measured, see table III, to determine compliance with the physical interchangeability requirement of 3.10. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) External and internal dimensions of cases, covers, and insertable subassemblies, when such dimensions affect mating of parts.

(b) Internal dimensions of cavities in cases and assemblies, when such dimensions affect acceptability of insertable subassemblies.

(c) Location of hinges and fasteners on separable parts or assemblies which must mate together, such as cases, covers, and mountings.

(d) Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts or plug-in assemblies and major components.

(e) Location of connectors, locking pins, fasteners, slides, and mountings on plug-in assemblies.

(f) Size and form of special threads.

4.10 Visual inspection of preparation for delivery. Packages and shipping containers shall be examined for the defects listed in Standard MIL–STD–252, as applicable and to determine conformance with the approved process sheet specified in 5.5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *Level A.* Ringer, Telephone TA–48/FT shall be preserved and packaged in accordance with the procedures specified for the designated methods as prescribed in Specification MIL–P–116 and described in the following paragraphs:

5.1.1.1 *Technical manual.* Each technical manual shall be packaged Method IC–3.

5.1.1.2 *Ringer, Telephone TA–48/FT.* Each telephone ringer shall be packaged Method IC–5, as follows: Cushion each telephone ringer on all surfaces with cells or pads or both, fabricated of double-faced, corrugated fiberboard, having a minimum dry bursting strength of 175 pounds, and designed to absorb the shock of impact encountered in handling and transit. The cells and pads shall be equal to the water-resistant requirements of the fiberboard box in which they are used. Place the cushioned telephone ringer within a close-fitting, fiberboard box, Style RSC, W5c, conforming to Specification PPP–B–636. Seal all seams and joints with waterproof, pressure-sensitive tape, Type III, Class I, olive drab, conforming to Specification PPP–T–60. Secure the technical manual to the top of the packaged telephone ringer with pressure-sensitive tape.

5.1.2 *Level C.* The ringer shall be packaged in accordance with the supplier's standard practice.

5.2 Packing.

5.2.1 *Level A.* Place four each packaged ringers and manuals in close-fitting boxes conforming to Specification PPP–B–621 (Class 2), PPP–B–601 (overseas type), PPP–B–585 (Class 3), or PPP–B–591 (overseas type). Closure and strapping shall be in accordance with the appendix to the box specification.

5.2.2 *Level B.* The ringer shall be packed as specified for Level A, except that boxes shall conform to Specification PPP–B–621 (Class 1), PPP–B–601 (domestic type), PPP–B–585 (Class 1), or PPP–B–591 (domestic type).

5.2.3 *Level C. Ringer,* Telephone TA–48/FT shall be packed for shipment in a manner conforming to the requirements of Uniform Freight Classification for rail ship-

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ment, National Motor Freight Classification for truck shipment, Parcel Post Regulations and the regulations of other carriers as applicable to the mode of transportation employed, at the lowest transportation rate.

5.3 **Marking.** Interior packages and exterior shipping containers shall be marked in accordance with the applicable provision of Standard MIL–STD–129.

5.4 **Package testing.** Packaging prepared as specified in 5.1.1 shall be required to pass the mandatory test requirements in Specification MIL–P–116, and the following rough handling test.

5.4.1 *Rough handling (pilot pack only).* Each completed pack (of the pilot pack, as prepared for shipment) shall be subjected to the rough handling test, as specified in Specification MIL–P–116. Packs shall be opened after testing and the packages and the items therein examined for damage. These items shall then be subjected to such operational tests as are necessary to determine freedom from malfunction.

5.5 **Pilot pack.** Unless otherwise specified in the invitation for bid or contract, a pilot pack shall be furnished for Government inspection and approval. A pilot pack shall consist of all preservation, packaging, packing, and marking for delivery for the complete equipment on contract or order. A Pilot Pack Process Sheet, using the format shown on figure 1, shall be prepared by the contractor and forwarded to the contracting officer and shall be accompanied by such drawings, sketches and other data as may be necessary to completely describe the procedure followed in fabricating the pilot pack and to identify the equipment items therein (see 6.2 (f) and (g)).

6. NOTES

6.1 **Intended use.** Ringer TA–48/FT is a telephone ringing equipment which converts a 60-cycle supply to a 20-cycle ringing current.

6.2 **Ordering data.** Requests, requisitions, schedules, and contracts or orders should specify the following:

(a) Title, number, and date of this specification.

(b) Directions for marking and shipping preproduction samples (see 3.2).

(c) Environmental and life test, samples and shipping required (see 3.2).

(d) Quantity of spare parts required (see 3.13).

(e) Leve of preservation and packaging and level of packing required (see 5.1 and 5.2).

(f) A packaging technician from the procuring agency will evaluate the pilot pack (see 5.5).

(g) Five copies of the process sheet and other pertinent data, as specified in 5.5, are required.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:
Army—Signal Corps
Navy—Bureau of Yards and Docks
Air Force

Other interest:
Navy—MC

Preparing activity:
Army—Signal Corps

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PACKAGING, PACKING, AND MARKING PILOT PACK PROCESS SHEET

CONTRACTOR ..... ORDER NO. .....
STREET ADDRESS ..... EQUIPMENT .....
CITY AND STATE ..... LEVEL .....

DATE PILOT PACK COMPLETED FOR EVALUATION

ITEM NAME							
PACKAGING METHOD							
CLEANING	SPECIFICATION						
	MATERIAL DESCRIPTION	T	G	C	T		G
PRESERVATION	SPECIFICATION						
	MATERIAL DESCRIPTION	T	G	C	T		G
CUSHIONING	SPECIFICATION						
	MATERIAL DESCRIPTION	T	G	C	T		G
	SIZE						
SECURING	SPECIFICATION						
	MATERIAL DESCRIPTION	T	G	C	T		G
	SIZE						
BARRIER	SPECIFICATION						
	MATERIAL DESCRIPTION	T	G	C	T		G
	SIZE						
DESICCANT	SPECIFICATION						
	MATERIAL DESCRIPTION	T	G	C	T		G
	SIZE						
UNIT CARTON	SPECIFICATION						
	MATERIAL DESCRIPTION	T	G	C	T		G
	SIZE						
INTERMEDIATE CARTON	SPECIFICATION						
	MATERIAL DESCRIPTION	T	G	C	T		G
	SIZE						
OUTER CARTON (CONSOLIDATED	SPECIFICATION						
	MATERIAL DESCRIPTION	T	G	C	T		G
	SIZE						
SECURE CLOSURE	SPECIFICATION						
	MATERIAL DESCRIPTION	T	G	C	T		G
	SIZE						

USE AS MANY COLUMNS AS NECESSARY

CASE LINER	SPECIFICATION	
	SEALING MATERIAL	
	SIZE	
SHIPPING CONTAINER	SPECIFICATION	
	STYLE	
	SIZE	
STRAPPING	SPECIFICATION	
	MATERIAL DESCRIPTION	
	SIZE	

LEGEND FOR MATERIAL: T — TYPE, G — GRADE, C — CLASS

FIGURE 1.

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